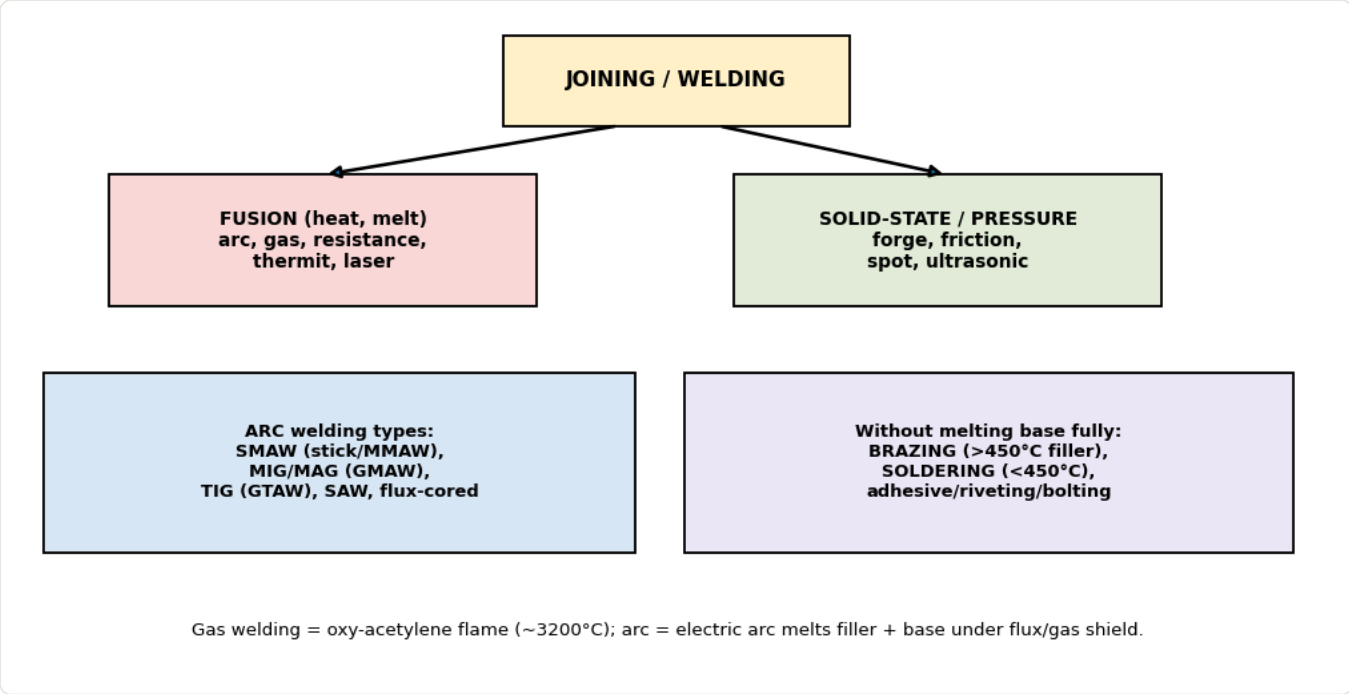
UNIT 8 · TOPIC 6(A)

Joining and Welding Processes

Syllabus topic: Basic joining processes: welding, weldment testing and metallurgy



Classification of joining and welding processes

Classification

Class	Processes
FUSION welding (joint melted)	ARC (electric arc — the workhorse); GAS (oxy-acetylene flame ~3200 °C); RESISTANCE (current + pressure — spot/seam); THERMIT; LASER/electron-beam
SOLID-STATE / PRESSURE welding (below melting point)	Forge, FRICTION, ultrasonic, explosive
Allied joining (base metal not fully melted)	BRAZING (filler melts above 450 °C, capillary action); SOLDERING (filler below 450 °C — electrical/sheet joints)
Mechanical / adhesive	Riveting, bolting, adhesives (Unit 2, Topic 2)

Common arc-welding types

Process	Electrode / shielding	Use
SMAW (shielded metal arc / stick / MMAW)	Flux-coated electrode	Cheap, portable — the mainstay of Indian farm-equipment fabrication
GMAW (MIG / MAG)	Continuous wire; inert (MIG) or active (MAG) gas	Fast, production work; DC reverse polarity gives some cleaning
GTAW (TIG)	Non-consumable TUNGSTEN electrode + separate filler; inert gas	High-quality welds on thin/non-ferrous metals; AC gives oxide cleaning on aluminium
SAW (submerged arc)	Granular flux blanket	Thick, long, automated welds

Shielding (gas or flux) protects the molten pool from atmospheric O_2 and N_2.

Practice MCQs

Q84. Welding that joins metals by melting the joint is:

(a) Pressure welding (b) Fusion welding
(c) Soldering (d) Riveting

Q85. Oxy-acetylene gas welding flame temperature is about:

(a) 800 °C (b) 1500 °C
(c) 3200 °C (d) 6000 °C

Q86. SMAW commonly refers to:

(a) TIG (b) Stick/manual metal arc welding with coated electrode
(c) MIG (d) Submerged arc

Q87. In TIG (GTAW) welding the electrode is:

(a) Consumable wire (b) Non-consumable tungsten
(c) Carbon rod (d) Flux tube

Q88. The shielding gas/flux in arc welding protects the pool from:

(a) Light (b) Atmospheric oxygen and nitrogen
(c) Water (d) Dust

Q89. Brazing uses a filler that melts:

(a) Below 450 °C (b) Above 450 °C but below base-metal melting
(c) Same as base metal (d) Never

Q90. Soldering filler melts at:

(a) Below 450 °C (b) Above 1000 °C
(c) 3200 °C (d) Same as steel

Q91. Resistance spot welding produces heat by:

(a) A flame (b) Electric current + pressure at the contact
(c) Friction (d) Chemical reaction

Answers

Q84 — (b) Fusion welding
Q85 — (c) 3200 °C
Q86 — (b) Stick/manual metal arc welding with coated electrode

Q87 — (b) Non-consumable tungsten

Q88 — (b) Atmospheric oxygen and nitrogen

Q89 — (b) Above 450 °C but below base-metal melting — *Brazing filler > 450 °C.*

Q90 — (a) Below 450 °C — *Soldering filler < 450 °C — the standard temperature line.*

Q91 — (b) Electric current + pressure at the contact

Sources: Hajra Choudhury — Elements of Workshop Technology (Vol I: casting, welding); Kalpakjian — Manufacturing Engineering; AWS welding references; literature on HAZ, weld defects, NDT and sand vs die casting (as given in the source notes, July 2026); NPTEL manufacturing lectures.